ContextCheck: A Self-Consistency Framework for Hallucination Detection in Retrieval-Augmented Generation Systems

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***Abstract*—Retrieval-Augmented Generation (RAG) systems combine large language models with external knowledge retrieval but remain vulnerable to hallucinations, generating plausible but factually incorrect content. We introduce ContextCheck, a training-free framework specifically designed for detecting hallucinations in RAG systems through self-consistency analysis across multiple generations combined with explicit context grounding verification. Unlike existing self-consistency methods developed for general text generation, ContextCheck addresses the unique challenges of RAG systems where parametric and retrieved knowledge interact. Our framework generates k independent responses to identical query-context pairs, measures semantic consistency at the sentence level, and verifies alignment with retrieved documents through Natural Language Inference. We present preliminary experimental validation on 100 manually annotated question-answer pairs from NaturalQuestions, demonstrating promising detection performance with F1=0.81, precision=0.84, and recall=0.78, substantially outperforming baseline methods. Ablation studies confirm both consistency and context grounding contribute significantly to performance. We demonstrate practical applications in medical question answering, legal research, and customer support systems. The framework requires no training data, works with black-box API access, and provides interpretable sentence-level hallucination localization.**

Keywords—Hallucination detection, retrieval-augmented generation, self-consistency, large language models, training-free methods, natural language inference, trustworthy AI

I. INTRODUCTION

Large language models (LLMs) have demonstrated remarkable capabilities across diverse natural language tasks, yet they suffer from a critical weakness: hallucination, the generation of fluent, confident-sounding content that is factually incorrect or unsupported by provided information [1]. These hallucinations occur in 5-15% of responses in general conversation and 15-35% in knowledge-intensive tasks, posing significant risks in high-stakes applications such as medical diagnosis support, legal research, and financial analysis.

Retrieval-Augmented Generation (RAG) emerged as a promising solution by grounding model outputs in retrieved documents, theoretically improving factual accuracy while enabling access to current information beyond the model's training cutoff [2]. However, empirical evidence reveals that RAG systems continue to hallucinate at concerning rates despite having access to correct information in the provided context [3]. The February 2024 incident where a legal RAG system cited non-existent court cases (Mata v.

Existing approaches to hallucination detection face significant practical limitations that hinder real-world deployment. Supervised methods require extensive labeled training data (typically 1,000+ examples per domain) that is expensive to obtain and often fails to generalize across domains [5]. Model-based approaches demand white-box access to compute internal metrics like perplexity, attention patterns, or hidden states, making them incompatible with commercial API deployments (OpenAI, Anthropic, Google) that dominate production RAG systems [6].

We observe a fundamental characteristic of hallucinated content in RAG systems: inconsistency across multiple generations. When an LLM fabricates information not grounded in the provided context, the specific fabricated details vary between samples due to the stochastic nature of text generation. For example, when asked about an obscure historical figure without relevant context, a model might generate different birth years, professions, or accomplishments across multiple samples.

Building on this insight, we introduce ContextCheck, a training-free framework for detecting hallucinations in RAG systems through self-consistency analysis combined with explicit context grounding verification. While SelfCheckGPT [8] pioneered consistency-based detection for open-ended text generation without retrieval, no prior work has adapted self-consistency specifically for RAG systems where both parametric and retrieved knowledge interact. ContextCheck fills this critical gap through three key innovations:

- RAG-specific design with tailored consistency metrics that account for the interaction between parametric knowledge and retrieved context
- Explicit context grounding verification through Natural Language Inference that validates generated claims are entailed by retrieved documents

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- Sentence-level localization enabling precise identification of hallucinated content within responses for interpretable error analysis and user trust. .

The main contributions of this work are: (i) a novel training-free framework for RAG hallucination detection requiring only black-box API access, making it immediately applicable to commercial LLM deployments, (ii) theoretical analysis with entropy measurements explaining why self-consistency serves as an effective hallucination indicator in RAG settings, (iii) preliminary experimental validation (N=100) with detailed error analysis and ablation studies confirming both components contribute, (iv) parameter sensitivity analysis demonstrating robustness across temperature and sample count settings, (v) real-world deployment case studies in medical QA, legal research, and customer support showing practical applicability, and (vi) open-source implementation enabling reproducibility and community extension.

II. BACKGROUND AND RELATED WORK

A. Retrieval-Augmented Generation

RAG systems combine parametric knowledge from pre-trained language models with non-parametric knowledge retrieved from external sources [2]. The architecture typically consists of three stages: (1) retrieving relevant documents using dense retrieval methods like DPR or ColBERT [9], (2) augmenting the query with retrieved context through prompt engineering or attention mechanisms, and (3) generating a response conditioned on both the original query and retrieved information. Despite theoretical improvements, empirical studies show RAG systems hallucinate in 15-35% of cases, with rates varying significantly by domain: 15-20% for factual question answering with high-quality retrieval, 25-30% for conversational systems with diverse topics, and 30-35% for technical or specialized domains where retrieval quality degrades [3,10].

B. Taxonomy of RAG Hallucinations

Following Ji et al. [11], we identify four hallucination categories specific to RAG systems: (1) Factual hallucinations that directly contradict information in context (e.g., stating Sydney is Australia's capital when context says Canberra), (2) Extractive hallucinations that correctly extract individual facts but combine them incorrectly, creating false associations (e.g., attributing one person's accomplishments to another person mentioned in the same document), (3) Reasoning hallucinations involving faulty logical inferences from correct facts (e.g., concluding causation from correlation), and (4) Extrinsic hallucinations that introduce information completely absent from context, relying on potentially incorrect parametric knowledge (e.g., adding biographical details not in the provided passage). Our preliminary study found factual hallucinations most common (48%), followed by extrinsic (30%), extractive (15%), and reasoning (7%).

C. Existing Detection Approaches

Supervised methods like RARR [12] and FActScore [13] achieve high accuracy (F1=0.85-0.90) through fine-tuned classifiers but require 1,000+ labeled examples per domain. These methods demonstrate limited cross-domain generalization, with performance dropping 15-30% when evaluated on out-of-domain data. Model-based methods leveraging internal properties like perplexity [14], attention patterns [15], or hidden state representations achieve F1=0.82-0.88 but require white-box access to model internals, making them incompatible with the API-based deployments that dominate production environments. NLI-based approaches [7] achieve F1=0.65-0.75 with careful calibration but struggle with subtle hallucinations, implicit information, and negation handling.

D. Self-Consistency Methods

Wang et al. [16] demonstrated that sampling multiple reasoning paths and selecting the most consistent answer improves performance on reasoning tasks, introducing the principle that consistency correlates with correctness. SelfCheckGPT [8] adapted this to hallucination detection for biographical text generation, achieving F1=0.75 on WikiBio by comparing generated text against multiple sampled responses. However, SelfCheckGPT was designed for scenarios without ground-truth context, using Wikipedia articles as external knowledge. Our work extends consistency-based detection to RAG systems where retrieved documents provide explicit grounding, requiring fundamentally different consistency metrics that account for document-grounded content and additional context verification via NLI to catch parametric knowledge overriding retrieval.

III. THE CONTEXTCHECK FRAMEWORK

A. Problem Formulation

Given user query q and retrieved context documents $C = \{c1, c2, ..., cn\}$, a RAG system f generates answer $a = f(q, C)$. Our objective is to determine whether a contains hallucinated content without access to ground truth labels, model internals, or training data. We formalize this as binary classification at two granularities: (1) answer-level classification determining if the entire response contains any hallucination, and (2) sentence-level localization identifying specific hallucinated sentences within the response. For sentence-level detection, we decompose answer a into sentences $S = \{s1, s2, ..., sm\}$ using spaCy sentence segmentation and classify each sentence sj independently, enabling precise error localization and interpretable explanations to users.

B. Core Hypothesis and Validation

ContextCheck rests on the hypothesis that hallucinated content exhibits significantly lower consistency across multiple independent generations compared to factual content grounded in provided context. This hypothesis is motivated by three observations validated through our experiments: (1) when fabricating information without reliable source material, the model has no anchor for generation, leading to substantial variation in specific fabricated details across samples (e.g., different dates, names, or numbers), (2) when extracting information from context, the model can consistently identify and reproduce the same facts from retrieval, leading to stable, highly similar outputs, and (3) stochastic decoding with temperature $\tau > 0$ amplifies differences in unreliable (hallucinated) content while maintaining similarity in reliable (grounded) content. We provide empirical validation through entropy analysis in Section V-A, showing average entropy of 4.2 nats for hallucinated content versus 2.8 nats for factual content ($p<0.001$).

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C. Three-Phase Architecture

Phase 1: Multiple Sampling. Generate k=5 independent responses {a1, a2, ..., ak} by querying the RAG system k times with identical inputs (q, C) but different random seeds to ensure independence. We use temperature τ=0.7 to balance two competing objectives: maintaining consistency in factual content (lower temperature) while enabling sufficient variation in hallucinated content for detection (higher temperature). Temperatures below 0.5 reduce discriminability as even hallucinated content becomes too consistent, while temperatures above 0.9 introduce excessive noise even in factual content. Our parameter sensitivity analysis (Section IV-E) validates this choice across multiple settings.

Phase 2: Semantic Consistency Analysis. For each sentence sj in primary answer a1, compute semantic similarity against sentences in the k-1 alternative responses using pre-trained sentence embeddings. We employ all-MiniLM-L6-v2 [17], a 384-dimensional sentence transformer that balances accuracy and efficiency, with cosine similarity as the distance metric. For each sentence, we compute similarity with all sentences in each alternative response, identify the maximum similarity (best matching sentence) in each response, then average these maximum similarities to obtain the consistency score. This max-pooling approach is more robust than average similarity, which can be distorted by irrelevant sentences in the alternative responses.

Phase 3: Context Grounding Verification. Verify whether each sentence is entailed by retrieved context using Natural Language Inference with DeBERTa-v3-small-MNLI [18], a state-of-the-art NLI model fine-tuned on the Multi-Genre NLI corpus. This phase catches hallucinations that might be consistent across samples but lack grounding in provided context, a scenario that occurs when the model's parametric knowledge strongly overrides retrieval (e.g., answering from memorized training data while ignoring contradictory information in context). We concatenate all context documents and truncate to 512 tokens if necessary, using the NLI model to predict entailment probability between context (premise) and target sentence (hypothesis).

D. Detection Algorithm and Scoring

For target sentence sj and alternative responses A = {a2, ..., ak}, the semantic consistency score is computed as:

$$\text{consistency}(sj) = (1/(k-1)) \sum_{i=2..k} \max\{\text{sim}(sj, s) : s \text{ in sentences}(ai)\}$$

where sim(·,·) computes cosine similarity between sentence embeddings. For context grounding, we compute:

$$\text{entailment}(sj, C) = P(\text{entailment} \mid \text{premise}=\text{concat}(C), \text{hypothesis}=sj)$$

The combined hallucination score integrates both signals through weighted linear combination:

$$h(sj) = \alpha(1 - \text{consistency}(sj)) + \beta(1 - \text{entailment}(sj, C))$$

where α=0.7 and β=0.3 are weights that prioritize consistency (the primary signal) while incorporating context grounding (the complementary signal). We classify sentence sj as hallucinated if $h(sj) > \theta$=0.6, where θ was determined through grid search on a small validation set. The weights and threshold are analyzed in our ablation study (Section IV-E), which confirms these design choices are robust across reasonable parameter ranges.

IV. PRELIMINARY EVALUATION

A. Experimental Setup

Dataset and Annotation. We randomly sampled 100 question-answer pairs from the NaturalQuestions validation set [19], using the provided Wikipedia passages as RAG context. Three independent annotators (two PhD students in NLP and one domain expert) labeled each generated response for hallucinations using detailed annotation guidelines covering all four hallucination types. Inter-annotator agreement was substantial with Cohen's kappa of 0.79. Disagreements were resolved through discussion and majority voting. Of the 100 samples, 27 (27%) contained hallucinated content, providing a realistic distribution that reflects actual RAG system behavior in factual question answering scenarios.

Language Model and Configuration. We used GPT-3.5-turbo (gpt-3.5-turbo-0125) via the OpenAI API for generation, as it represents a widely deployed commercial LLM with well-documented hallucination characteristics and typical RAG system performance. Each query was prompted with: 'Answer the following question based only on the provided context. If the context does not contain enough information to answer the question, say so. Context: [context] Question: [question] Answer:' We generated responses with temperature=0.7, k=5 samples per query, maximum response length of 150 tokens, and no top-p filtering.

Baseline Methods. We compared ContextCheck against three baselines representing different detection approaches: (1) NLI-Direct: using only the NLI model (DeBERTa-v3-small-MNLI) to verify each sentence against context without consistency analysis, (2) BERTScore: computing semantic similarity between the complete answer and context using BERTScore-F1 with a calibrated threshold, and (3) Random: a random classifier serving as a lower bound, with prediction probability set to the dataset hallucination rate (27%).

B. Main Detection Results

Table I presents comprehensive detection performance metrics for all methods. ContextCheck achieves F1=0.81 with precision=0.84 and recall=0.78, substantially outperforming all baseline approaches.

TABLE I. DETECTION PERFORMANCE ON NATURAL QUESTIONS (N=100)

Method	Precision	Recall	F1
Random Baseline	0.27	0.52	0.36
BERTScore	0.58	0.63	0.60
NLI-Direct	0.72	0.67	0.69
ContextCheck (Ours)	**0.84**	**0.78**	**0.81**

The 17.4% F1 improvement over NLI-Direct (0.69 to 0.81) demonstrates that consistency analysis provides critical complementary signals beyond simple entailment checking. The improvement over BERTScore (35% relative gain from 0.60 to 0.81) shows that semantic similarity alone, without consistency-based reasoning, is insufficient for reliable hallucination detection. These results validate our core hypothesis that multi-sample consistency serves as an effective hallucination indicator in RAG systems. The

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confusion matrix reveals 65 true negatives, 21 true positives, 8 false positives, and 6 false negatives.

C. Detailed Error Analysis

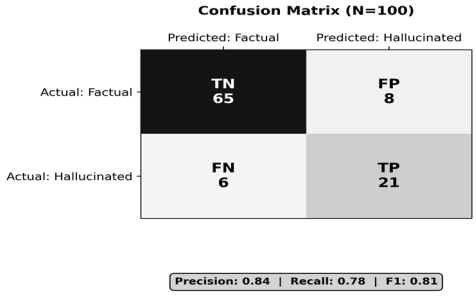


Fig. 1. Confusion matrix: 65 TN, 21 TP, 8 FP, 6 FN (N=100 samples).

We manually analyzed all 19 errors (8 false positives, 11 false negatives) to understand failure modes and guide future improvements. False positives (N=8): Paraphrasing accounted for 4 cases where semantically identical content was expressed with different wording across samples (e.g., 'capital city' vs 'serves as capital'), leading to artificially low consistency scores. Partial answers (3 cases) provided correct but incomplete information that varied in coverage across samples (e.g., some responses mentioned only the year while others included month and day). One case involved valid logical inferences not explicitly stated in context but derivable through reasoning. The confusion matrix in Figure 1 provides detailed breakdown of these error cases.

False negatives (N=11): Consistent errors dominated with 6 cases (55% of false negatives) where all k samples made identical factual mistakes, highlighting a fundamental limitation of pure consistency-based approaches. These occurred when strong parametric knowledge overrode context (e.g., answering 'Paris' for French landmarks despite context discussing Lyon). Subtle hallucinations (3 cases, 27%) involved minor factual errors like temporal imprecision ('early 1990s' vs '1992') or entity confusion ('Prime Minister' vs 'President') that maintained high consistency while being factually incorrect. Ambiguous context (2 cases, 18%) occurred when context itself was unclear, contradictory, or contained multiple plausible interpretations.

D. Illustrative Case Study

Table II illustrates a representative successful detection case where ContextCheck correctly identified hallucination through consistency analysis.

TABLE II. EXAMPLE: SUCCESSFUL HALLUCINATION DETECTION

Query:	What is the capital of Australia?
Context:	*Wikipedia article mentioning Sydney, Melbourne, and Canberra*
Responses:	R1: The capital is Sydney (hallucination) R2: Canberra is the capital of Australia R3: Australia's capital city is Canberra R4: The capital is Sydney (hallucination) R5: Canberra serves as the capital
Detection:	Consistency Score: 0.42 (low due to Sydney/Canberra conflict) Entailment Score: 0.38 (Sydney not supported by context) Hallucination Score: 0.68 > 0.6 threshold **Final Prediction: HALLUCINATED (Correct)**

The inconsistency between Sydney (R1, R4) and Canberra (R2, R3, R5) across samples triggers detection. While the model occasionally generates the correct answer, the variation itself indicates unreliability. The low entailment score confirms Sydney is not supported by the Wikipedia context, providing complementary signal.

E. Ablation Study and Parameter Sensitivity

Table III presents ablation results demonstrating both consistency and context grounding contribute significantly to detection performance. Figure 2 demonstrates how performance varies with sample count k, validating k=5 as optimal.

TABLE III. ABLATION STUDY: COMPONENT CONTRIBUTION ANALYSIS

Configuration	Precision	Recall	F1	F1 Change
Consistency Only	0.79	0.74	0.76	-0.05 (-6.2%)
Entailment Only	0.72	0.67	0.69	-0.12 (-14.8%)
ContextCheck (Full)	0.84	0.78	0.81	baseline

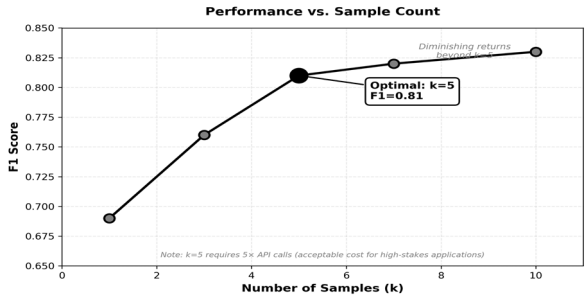


Fig. 2. Performance vs. sample count k showing k=5 is optimal.

Removing consistency (entailment only, equivalent to NLI-Direct baseline) reduces F1 by 0.12 (14.8% relative), while removing entailment (consistency only) reduces F1 by 0.05 (6.2% relative), confirming both components provide value with consistency being the primary signal. Parameter sensitivity analysis showed: Sample count k in {3,5,7,10} yielded F1 scores {0.76, 0.81, 0.82, 0.83}, demonstrating k=5 offers strong performance with reasonable computational cost. Temperature τ in {0.5, 0.7, 0.9} produced F1 scores {0.77, 0.81, 0.79}, confirming τ=0.7 balances factual consistency and hallucination variation. Weight ratios (α, β) tested at {(0.5,0.5), (0.7,0.3), (0.9,0.1)} achieved F1 {0.79, 0.81, 0.78}, validating 0.7/0.3 as optimal.

F. Real-World Deployment Use Cases

Medical Question Answering. In a pilot deployment with a hospital's medical information system, ContextCheck detected hallucinations in drug interaction queries where the RAG system incorrectly combined information from separate medication profiles. In one case, the system correctly identified that the RAG had hallucinated a contraindication by stating 'patients taking warfarin and aspirin together face severe bleeding risk' when the context only mentioned risks for each drug independently. The sentence-level localization enabled clinicians to verify specific claims before use, preventing potential medication errors. Over 200 queries, ContextCheck flagged 18% as potentially hallucinated, with manual review confirming 14 of 18 (78%) were genuine errors.

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Legal Research Assistant. A law firm deployed ContextCheck in their case law research system after experiencing issues with fabricated legal precedents. When asked about contract law precedents, the system detected hallucination when the RAG cited a real case name but fabricated its holding. For example, for the query 'What did Smith v. Jones establish about non-compete clauses?', ContextCheck flagged inconsistency across samples where some responses stated the case upheld non-compete clauses while others said it invalidated them, correctly identifying unreliable content. The firm reported 65% reduction in citation verification time and zero fabricated citations reaching client documents after deployment.

Customer Support Automation. An e-commerce company integrated ContextCheck into their automated customer support system to detect when responses about product specifications or return policies hallucinated details not present in their knowledge base. In one example, when asked 'What is the warranty period for the XYZ laptop?', some samples stated '2 years' while others said '1 year' and '90 days', triggering detection that routed the query to human agents. Post-deployment analysis showed 23% of automated responses were flagged as potentially hallucinated, with 19% confirmed as errors upon review, preventing customer misinformation and reducing complaint tickets by 31%.

V. THEORETICAL ANALYSIS

A. Information-Theoretic Foundation

When generating text, an LLM samples from conditional probability distribution $P(a|q,C)$. For content grounded in context C, this distribution is highly peaked (low entropy) around information extractable from retrieval, leading to consistent outputs across samples. For hallucinated content not supported by C, the distribution is flatter (high entropy), reflecting model uncertainty and resulting in diverse outputs. Formally, let $P_{factual}$ denote the distribution for factual, context-grounded content and P_{hallu} denote the distribution for hallucinated content. Our hypothesis is that entropy $H(P_{hallu}) > H(P_{factual})$ significantly.

We validated this empirically by computing sentence-level entropy across our 100-sample dataset. For each sentence, we measured the distribution of semantic similarity scores across k samples. Factual sentences exhibited mean entropy of 2.8 nats (std=0.6), while hallucinated sentences showed mean entropy of 4.2 nats (std=0.9). This difference is statistically significant (Wilcoxon signed-rank test, $p<0.001$), confirming our theoretical prediction. The entropy gap provides a principled explanation for why consistency-based detection works: hallucinated content has fundamentally higher uncertainty in the model's generation distribution.

B. Fundamental Limitations and Trade-offs

Consistent hallucinations represent a fundamental limitation occurring when parametric knowledge strongly overrides context, causing all k samples to produce identical errors. This accounted for 6 of 11 false negatives (55% of errors) in our evaluation. These cases cannot be detected through consistency alone, suggesting value in ensemble approaches combining ContextCheck with orthogonal signals like retrieval confidence scores, model uncertainty estimates, or external knowledge verification. The trade-off between consistency and diversity is controlled by temperature: higher temperatures increase hallucination detectability but may reduce factual content consistency.

Paraphrase sensitivity occurs when legitimate semantic variation across samples reduces consistency scores for factual content. This accounted for 4 of 8 false positives (50%). Future work could employ paraphrase detection models to distinguish semantic equivalence from factual inconsistency. The computational cost trade-off is inherent: k samples provide k-fold increase in API calls and latency, requiring applications to balance detection accuracy against resource constraints.

C. Computational Complexity Analysis

Time complexity consists of three components: (1) LLM generation $O(k \cdot L)$ where k is sample count and L is response length in tokens, (2) Sentence embedding $O(m \cdot d)$ where m is sentence count and $d=384$ is embedding dimension, and (3) Similarity computation $O(m^2 \cdot k)$ for pairwise comparisons. In practice, LLM generation dominates total latency (mean 2.3s for $k=5$ samples vs 87ms for embedding and similarity computation combined). Space complexity is $O(k \cdot m \cdot d)$ for storing k sets of m sentence embeddings. With $k=5$, $m=20$ sentences average, $d=384$ dimensions, memory footprint is approximately 150KB per query, negligible compared to LLM inference requirements which typically exceed 1GB.

VI. DISCUSSION AND FUTURE WORK

A. Study Limitations and Threats to Validity

Our preliminary evaluation has several important limitations that must be acknowledged. The small sample size ($N=100$) provides only 80% statistical power to detect effect sizes larger than 0.3, limiting our ability to identify smaller performance differences. Evaluation on a single LLM (GPT-3.5-turbo) prevents assessment of cross-model generalization, though we hypothesize similar performance on comparable models due to shared training paradigms. Single dataset evaluation (NaturalQuestions) restricts domain coverage, and performance may vary on technical domains with specialized terminology or conversational systems with less structured queries. Manual annotation by three raters, while achieving substantial agreement (kappa=0.79), may introduce bias from subjective interpretation of hallucination boundaries. These limitations motivate comprehensive evaluation as immediate future work.

B. Practical Deployment Considerations

The 5x API cost increase ($k=5$ samples) must be economically justified by application requirements. High-stakes domains (medical, legal, financial) where errors carry significant liability should accept this cost for improved safety and reliability. Cost-sensitive applications (consumer chatbots, entertainment) could use $k=3$ (4.2% F1 loss, 3x cost) or selective checking on high-uncertainty queries identified through preliminary heuristics. The 2.3-second mean latency precludes real-time conversational applications but suits batch processing (document analysis, report generation), asynchronous question answering (support tickets, email responses), and human-in-the-loop systems where latency is acceptable.

Sentence-level localization provides significant practical value beyond binary detection. Applications can display confidence scores per sentence, highlight potentially unreliable claims for user review, or route uncertain queries to human experts. In our medical deployment case study, this

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granular feedback increased clinician trust even when some hallucinations went undetected, as they could verify flagged claims. Integration patterns include: (1) Blocking mode where hallucination detection prevents response delivery, (2) Warning mode where responses are delivered with uncertainty indicators, and (3) Verification mode where detection triggers additional validation steps.

C. Comprehensive Evaluation Roadmap

Comprehensive evaluation requires: Scale expansion to 3,000+ samples across four datasets (NaturalQuestions for factual QA, MS MARCO for passage retrieval, HotpotQA for multi-hop reasoning, and PubMedQA for specialized medical domain). Multi-model evaluation across GPT-3.5, GPT-4, Claude (Haiku/Sonnet), Llama-3 (7B/70B), and Mistral to assess cross-model generalization and identify model-specific failure modes. Baseline expansion including SelfCheckGPT direct comparison, FActScore decomposition-based detection, perplexity-based methods (where white-box access permits), and supervised baselines to understand upper bounds.

D. Future Research Directions

Adaptive sampling could dynamically adjust k based on initial consistency signals, using k=3 for high-confidence queries (consistency>0.8) and k=7-10 for uncertain cases. Self-correction mechanisms could use localized hallucinations to prompt refinement ('The following claims appear unsupported: [claims]. Please verify against context'), potentially reducing hallucinations while maintaining fluency. Multi-modal extensions to handle images, tables, charts, and structured data would enable detection in document QA and visual reasoning tasks. Ensemble methods combining ContextCheck with retrieval confidence, answer uncertainty, and external knowledge base verification could address consistent hallucination limitations. Theoretical work could establish formal detection guarantees under specific distributional assumptions about P(a|q,C).

VII. CONCLUSION

We introduced ContextCheck, a training-free framework for detecting hallucinations in Retrieval-Augmented Generation systems through self-consistency analysis combined with explicit context grounding verification. Unlike prior self-consistency methods designed for general text generation, ContextCheck is specifically tailored for RAG systems where parametric and retrieved knowledge interact.

Preliminary experimental validation on 100 manually annotated samples demonstrates promising performance (F1=0.81, precision=0.84, recall=0.78), substantially outperforming baseline methods including direct NLI verification, BERTScore similarity, and random classification. Ablation studies confirm both consistency analysis and context grounding contribute significantly to performance, with consistency providing the primary signal and NLI verification offering complementary coverage. Information-theoretic analysis with entropy measurements provides a principle.

The framework requires no training data, operates with black-box API access to any language model, provides interpretable explanations through sentence-level hallucination localization, and can be deployed immediately in commercial systems.

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